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[141. Linked List Cycle]

Easy

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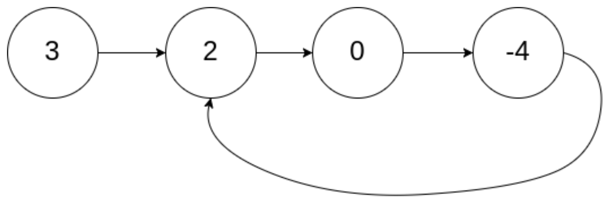
Given `head`, the head of a linked list, determine if the linked list has a cycle in it.

There is a cycle in a linked list if there is some node in the list that can be reached again by continuously following the `next` pointer. Internally, `pos` is used to denote the index of the node that tail's `next` pointer is connected to. **Note that `pos` is not passed as a parameter**.

Return `true` *if there is a cycle in the linked list*. Otherwise, return `false`.

Example 1:

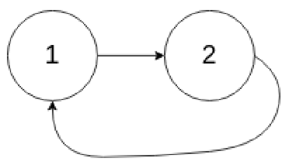


Input: head = [3,2,0,-4], pos = 1

Output: true

Explanation: There is a cycle in the linked list, where the tail connects to the 1st node (0-indexed).

Example 2:



Input: head = [1,2], pos = 0

Output: true

Explanation: There is a cycle in the linked list, where the tail connects to the 0th node.

Example 3:

Input: head = [1], pos = -1

Output: false

Explanation: There is no cycle in the linked list.

Constraints:

- The number of the nodes in the list is in the range `[0, 10^4]`.
- `-10^5 <= Node.val <= 10^5`
- `pos` is `-1` or a **valid index** in the linked-list.

Follow up: Can you solve it using `O(1)` (i.e. constant) memory?

Seen this question in a real interview before?

1/6

Yes

No

Accepted

4,989,039/9.2M

Acceptance Rate

54.5%

Companies

Discussion (533)

💡 Discussion Rules

1. Please don't post **any solutions** in this discussion.

2. The problem discussion is for asking questions about the problem or for sharing tips - anything except for solutions.

3. If you'd like to share your solution for feedback and ideas, please head to the solutions tab and post it there.

Andrey

Aug 05, 2023

terrible description

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1.3K

5

Evan Wang

Jul 13, 2019

I don't see why we need 'pos' in the inputs, while we don't see it in the code. It's kind confusing.

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1.3K

17

[Aymen Sekhri]
Aug 29, 2020

I'm confused why `pos` is a input and not in argument list ?

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12

justdvl
Jun 25, 2021

Can someone please explain this task? Not solution, I don't understand the task itself.

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sdhru
Nov 27, 2022

If we iterate through a linked list in conventional way(head = head.next) . We need to find if the next pointer of the last node points to null value or to a prepresent node in the list

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[amanuelgaromsa]

Dec 27, 2022

[@sdhru](#) how can we terminate the traverse if the last node is still pointing to another node?

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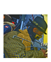

[Saikiran Prodaturu](#)

Mar 28, 2023

[@amanuelgaromsa If the last node is pointing to another node, it means we have a cycle, which we had to find. Include a condition like the pos is already visited then break the loop and return based on the position you are at.

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1

[Carlos Granados](#)

Apr 23, 2023

[@sdhru](#) The only way I know I've reached the end of a list is when next is null, but if it's never null, then how do you know you reached the end?

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[GustavoWilliam](#)
Apr 26, 2023

@sprodaturu To check if the pos has been visited you would have to store all visited nodes in a data structure and therefore the space complexity would not be constant as the task is demanding.

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AbstractedAway

Jul 30, 2023

Visualization of the Tortoise and Hare algorithm

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Cyclic8222
Dec 11, 2023

This is a knowledge based question.

If you've never heard of `Floyd’s cycle-finding algorithm` or the `fast-slow method`, you're unlikely to figure it out yourself in 20 mins.

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18

Akash Yadav

Dec 30, 2023

I couldn't solve it in 20 mins but I did it without knowing Floyd's cycle, it's a tricky question but if someone knows/have solved questions of visiting numbers in array to mark duplicates then it could be easier.

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ategale

Feb 29, 2024

A set should be similar performance wise (besides memory usage).

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webwraith
Mar 05, 2024

I disagree with the whole "unlikely to figure it out in 20 minutes" part. As has already been stated, a set (or in my case, an unordered_set) of the pointers, and checking the sets size after each insertion came to mind quite quickly.

Then again, maybe I just don't think like most people.

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Prashant Singh

Apr 07, 2024

Took me 1 min to solve just taken list and added each node whenever i encountered the same node break;

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vinyx

May 15, 2024

@Prashantishere thats O(n) space complexity. Try doing it in O(1)

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[Piyush](#)

Sep 04, 2023

This is the first question I solved on Leetcode, 2 years back. I guess my life's also a cycle :)

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3

jasonbayk
Oct 15, 2019

The "pos" should not be a part of the input, in fact it is not known a priori. Should be removed for better clarity.

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1

vicky

Jul 13, 2021

We can prove it using relative velocity kind of thing.

If there is a cycle then there will be a time when walker will enter the cycle for the first time.

At that moment runner will already be present in the cycle(because he was fast and ahead).

Now if the runner is also on the same node as walker problem solved.

Otherwise If both fast and slow are on different nodes on cycle then relative velocity of runner with respect to walker is one node per iteration (because walker moves one step then runner moves two step therefore relative velocity is 2 - 1 = 1 it can we viewed as runner is closing in on walker). So that means runner will surely catch the walker.

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7

anon301176

Feb 14, 2023

my solution is failing under this test case: [-21,10,17,8,4,26,5,35,33,-7,-16,27,-12,6,29,-12,5,9,20,14,14,2,13,-24,21,23,-21,5] very clearly there is a loop in this data, but its telling me i should be returning false. can someone explain this?

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```
 * }
 *     }
 *         next = null;
 *         val = x;
 *     ListNode(int x) {
 */
public class Solution {
    public boolean hasCycle(ListNode head) {
```

}

*   ListNode next;

* class ListNode {

*   int val;

}

Saved

Ln 4, Col 14

head =

[3,2,0,-4]

pos =

1

9

1

2

3

4

5

6

›

[3,2,0,-4]

1

[1,2]

0

[1]

-1

</>
Source

?

You must run your code first

😎

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